# FIX MY FORMAT, AGENT PAPERPAL

*Build an AI agent that does what no tool has ever done: automate academic manuscript formatting end-to-end.*

**HackaMined 2026**

Paperpal

Cactus Communications | HackaMined 2026

www.paperpal.com

THE CONTEXT

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# Why Does "Formatting" Even Matter?

## What "formatting" covers

- Citation style — APA, Vancouver, IEEE, Chicago…
- Heading hierarchy, abstract structure, word limits
- Figure placement, captions, statistical notation
- Reference list ordering & punctuation

## Why journals insist on it

- Keeps peer reviewers focused on science, not layout
- Enables consistent indexing & citation traceability
- Signals author credibility — price of entry

**46,700**

**active peer-reviewed journals**

*each with unique rules — no two identical*

*Source: STM Global Report*

- Rules change per issue, section, and article type
- No open, machine-readable database of these rules exists
- Special issues add entirely new formatting requirements

THE PROBLEM

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# The Scale of the Problem

*Academic formatting — a hidden, measurable tax on every researcher's time*

**30–70%**

**of manuscripts desk rejected**

*Before peer review begins (Elsevier: 30–50%; top journals: 90%+)*

**30+ hrs**

**per manuscript spent reformatting**

*Editage global survey of 7,000+ researchers across disciplines*

**$1B+/yr**

**annual reformatting burden globally**

*Billions of researcher-hours lost every year to manual re-formatting*

**4M+**

**papers published every year**

*Every single one needs manual formatting before submission*

*Sources: Editage Global Author Survey · Elsevier desk rejection data · STM Global Report · AJE Author Insights*

THE HARD PART

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# Why Hasn't This Been Solved?

*Existing tools are partial fixes.*

### No Rule Database

Journal guidelines live in PDFs and prose pages — no API, no schema. An AI must extract and re-interpret them fresh every time.

### Documents Are Multimodal

Real manuscripts mix text, embedded figures, tables, equations and footnotes. Structure detection across all of this is deep NLP — not regex.

### Privacy Is a Hard Wall

Unpublished research can't go to generic LLM APIs — IP risk, patent novelty, export controls. Researchers simply won't upload.

### Consistency Is Global

Change one citation style and every in-text reference, every reference list entry, every cross-reference must stay in sync. One error breaks traceability.

### No Training Ground Truth

No public dataset of 'messy manuscript → correctly formatted' pairs exists. You can't fine-tune — you must build a rule interpreter.

### Output Must Be Publication-Ready

The result isn't a suggestion — it's a transformed .docx that opens in Word, preserves every figure and table, and is directly submittable.

THE CHALLENGE

# Build an Agentic Manuscript Formatter

*Take a research manuscript in any format and transform it to any journal's specifications — autonomously.*

**01 Ingest**
Word / PDF or plain text

→

**02 Parse**
Detect all structural elements

→

**03 Interpret**
Read journal formatting rules

→

**04 Transform**
Reformat the whole document

→

**05 Validate**
Check citation–reference consistency

***Think: agentic rule interpreter, not a template applier.***

**Input:** Any manuscript (Word/PDF) **Output:** Reformatted, validated .docx or LaTeX ⚡ **Partial steps evaluated with equal weightage**

INNOVATION DIRECTIONS

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# Push the Boundaries

### LLM Rule Extraction

Parse any journal's 'Instructions for Authors' page live — no pre-built rule database needed

### CSL + NLP Hybrid

CSL covers 10,000+ journal citation styles. Layer structural rules on top — headings, margins, abstracts

### Citation Consistency Engine

Every in-text cite resolves to exactly one reference entry — and vice versa — across the full document

### Modular Reformatting

Reformat one section without touching the rest — citations only, headings only, figures only

### Explainable Corrections

Show what changed and why — 'Heading changed to H3 per APA 7th §2.27' builds researcher trust

### Multi-Agent Architecture

Separate agents for rule extraction, document parsing, transformation, and validation — composable pipeline

# PRIZES & MENTORS

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**Winner**

**INR 15K**

**1st Runner Up**

**INR 10K**

***+ AI Engineering Internship opportunity!***

## YOUR MENTORS

**Riddhi Shah**

Product Manager

**Sachinkumar Patel**

Software Architect

**Kunal Sawhney**

Sr. Manager, Engineering

**Fenil Ramoliya**

ML Engineer (MiNED '24) 🏆

**Bhargav Modha**

ML Engineer (MiNED '23) 🏆